

AI Study Preparation & Paper Prediction Agent

Simple Problem Formulation and Environment Classification

1. Problem Formulation

Element	Description
Initial State	The agent receives study guides, lecture notes, past papers, and the course syllabus. It has no prior assumption about which topics are most important.
Actions	Read and organize material; extract topics and question types; compare past papers; identify recurring patterns; rank topics; generate a study plan and practice questions.
Transition Model	When the agent analyzes new material or feedback, its knowledge base, topic rankings, confidence levels, and study recommendations are updated.
Path Cost	The cost may be measured by study time, number of topics reviewed, or effort required. The objective is to use limited preparation time efficiently.
Goal Test	The student receives a clear, evidence-based priority list and a practical revision plan, with explanations for each recommendation.
State Space	All possible combinations of uploaded materials, extracted topics, question patterns, student progress, confidence scores, and generated study plans.
Search Space	All possible sequences of analyzing material, comparing patterns, ranking topics, generating recommendations, receiving feedback, and updating the plan.
Agents	Single agent. One intelligent agent supports the student; there is no competing or cooperating agent in the defined task.

2. Environmental Properties

Property	Classification	Why
Observability	Partially Observable	The agent only knows the material it receives. It cannot see the actual exam or the examiner's private intentions.
Determinism	Stochastic / Non-deterministic	Past patterns can suggest likely topics, but they cannot guarantee what will appear. Unexpected questions may be included.
Episodes	Sequential	Earlier study recommendations and student performance affect later recommendations.
Change	Dynamic	New notes, past papers, and quiz results may arrive and change the knowledge base.
Values	Discrete & Continuous	Topics and question types are discrete, while frequency, confidence, and probability estimates can be continuous.
Agents	Single Agent	One intelligent agent assists one student.

3. Choosing the State Type

Option	Suitability	Explanation
Multi-state	Most suitable	The agent moves through many possible states: raw documents, extracted topics, detected patterns, ranked topics, study plans, and updated plans.

Final Answer: The AI Study Preparation & Paper Prediction Agent is best classified as a **Multi-state agent** because it passes through multiple changing states while processing study material, detecting patterns, creating recommendations, and updating its plan using feedback. It is also partially observable and stochastic because it cannot access the actual exam or guarantee exact questions.